HMM analysis of power usage data

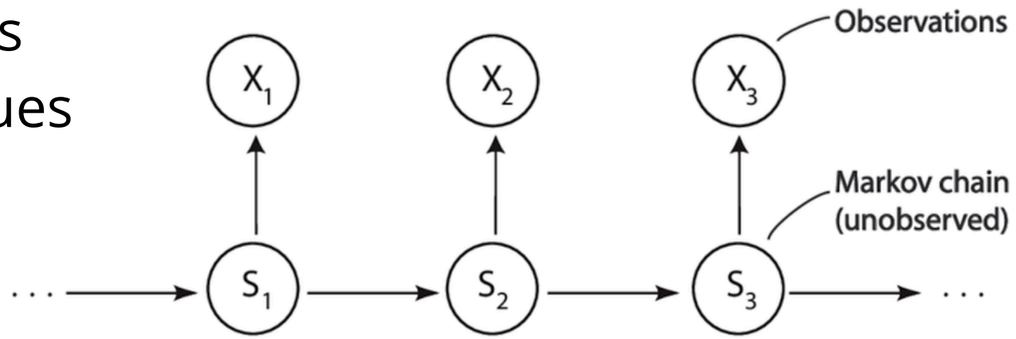
by Kateryna Rozenko and Lucia Majerčáková



Introduction

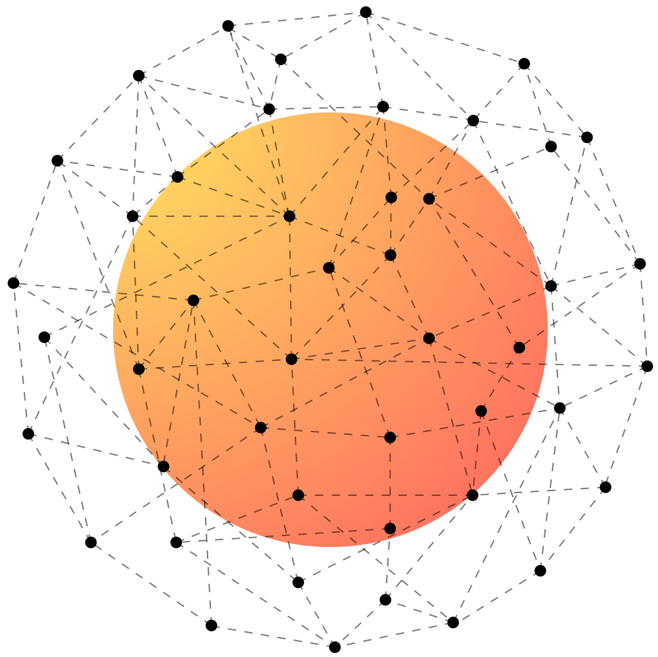
Data:
Time-series data of power usage in a household in Houston, Texas, USA.

Problem:
Using Hidden Markov Models to find underlying states and distributions → can be used to predict future values



https://www.kaggle.com/datasets/srinuti/residential-power-usage-3years-data-timeseries?select=weather_2016_2020_daily.csv

Table of Contents



Dataset & Data Processing 4

Simple HMMs (Differing # of States) 8

- Model Selection & Checking 10

Models with Covariates 13

Overall Model Selection 19

State Decoding 21

Results & Reflections 24

Dataset



One household, ~35k datapoints, hourly data

Variables:

start_datetime: 2016-01-06 00:00:00 - 2020-07-07 23:00:00
value_kWh: 0.00+
month_of_year: 1-12
hour_of_day: 0-23
day_of_week: 0-6 (Mon-Sun)
notes: "weekday" "weekend" "vacation" "COVID_lockdown"

```
   value_kwh
Min.   :0.0640
1st Qu.:0.3237
Median :0.5010
Mean   :0.8916
3rd Qu.:1.0720
Max.   :6.4460
```

	start_datetime	value_kWh	day_of_week	notes	month_of_year	hour_of_day
35927	2020-07-06 22:00:00	2.156	6	COVID_lockdown	7	22
35928	2020-07-06 23:00:00	1.492	6	COVID_lockdown	7	23
35929	2020-07-07 00:00:00	1.601	1	weekday	7	0
35930	2020-07-07 01:00:00	1.147	1	weekday	7	1
35931	2020-07-07 02:00:00	1.992	1	weekday	7	2
35932	2020-07-07 03:00:00	1.104	1	weekday	7	3
35933	2020-07-07 04:00:00	1.072	1	weekday	7	4
35934	2020-07-07 05:00:00	0.940	1	weekday	7	5

Data preparation

	StartDate	Value (kWh)	day_of_week
283	2016-12-06 19:00:00	1.156	1
284	2016-12-06 20:00:00	1.306	1
285	2016-12-06 21:00:00	2.927	1
286	2016-12-06 22:00:00	3.135	1
287	2016-12-06 23:00:00	1.454	1
288	2016-06-13 00:00:00	0.956	0
289	2016-06-13 01:00:00	1.290	0
290	2016-06-13 02:00:00	1.182	0
291	2016-06-13 03:00:00	0.909	0

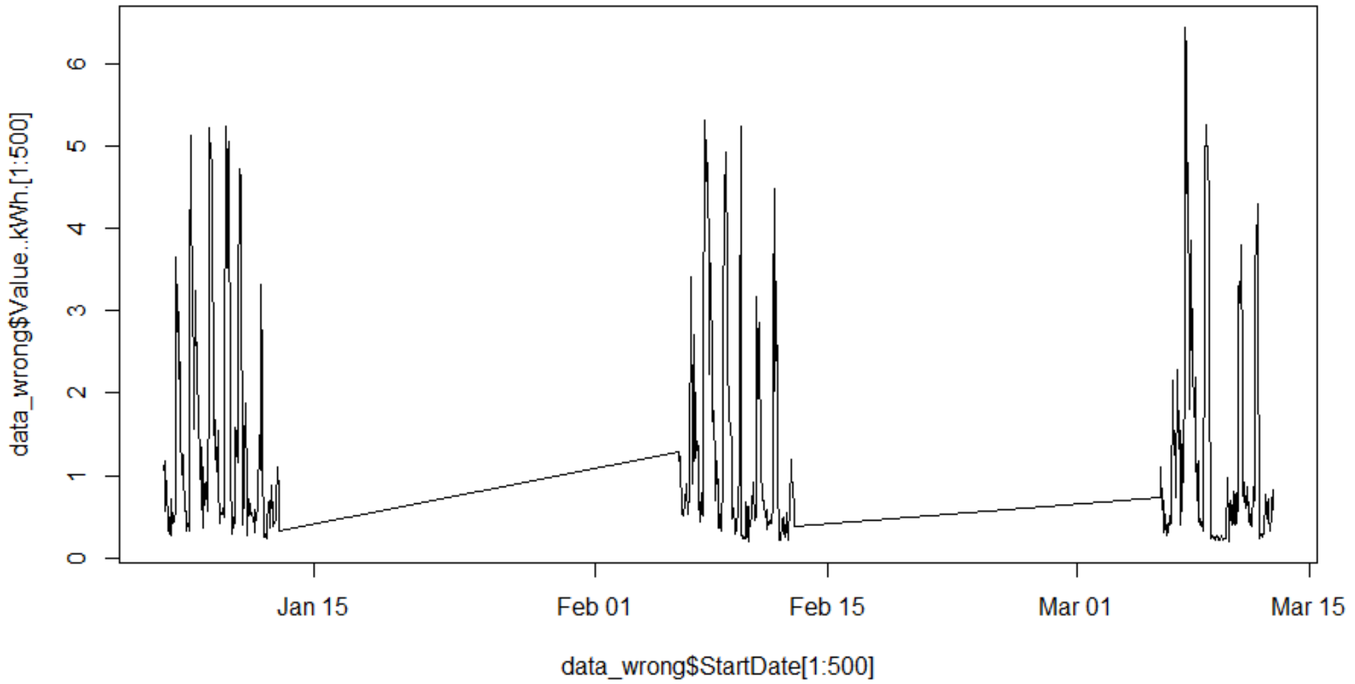
Date format was consistently inconsistent → **fixed**

- *"day_of_week"* and *"weekday", "weekend"* in *"notes"* based on the wrong old format → partially **distorted information**

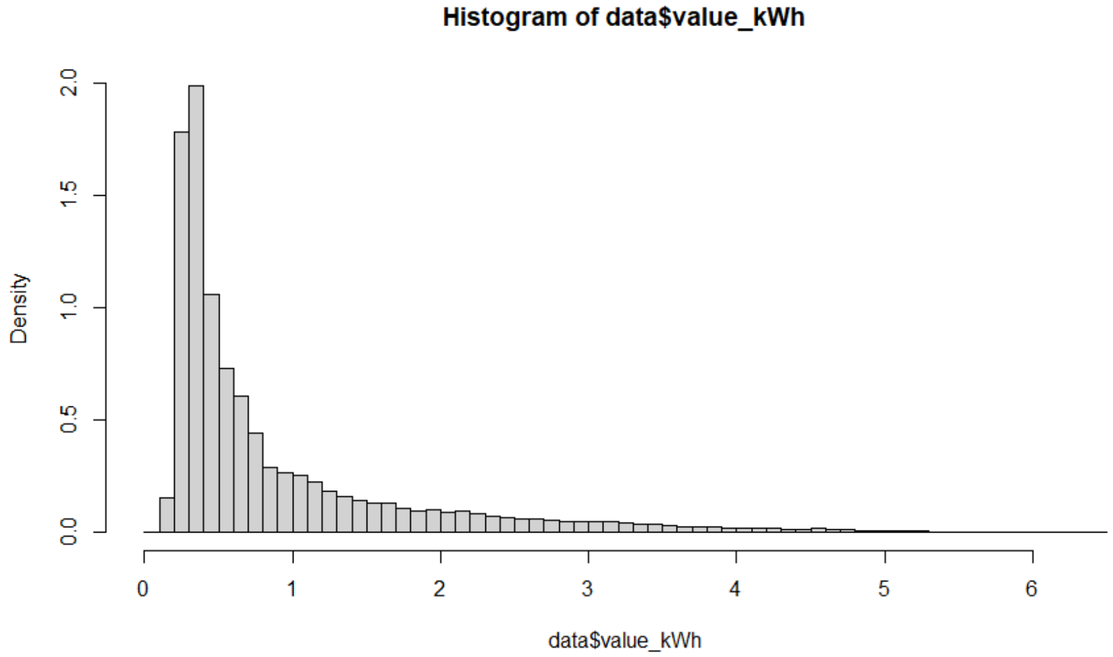
Added variables: "time_of_day", "month_of_year"

Issue with summer time changes
→ **UTC format** to not have duplicate and missing values

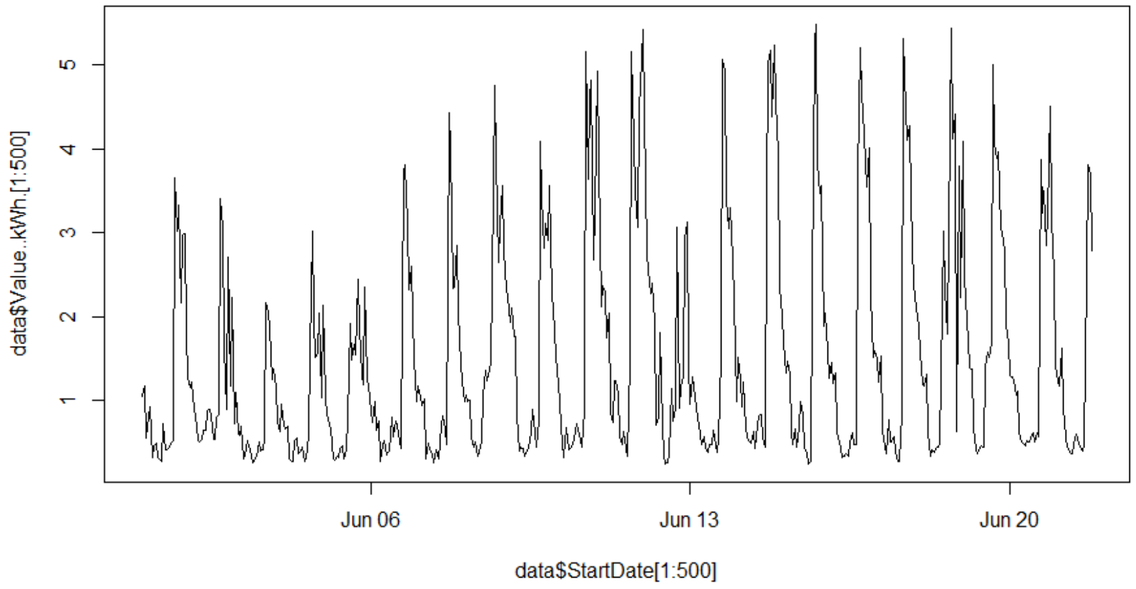
as.POSIXct(data$start_datetime, format="%Y-%m-%d %H:%M:%S", tz="UTC")

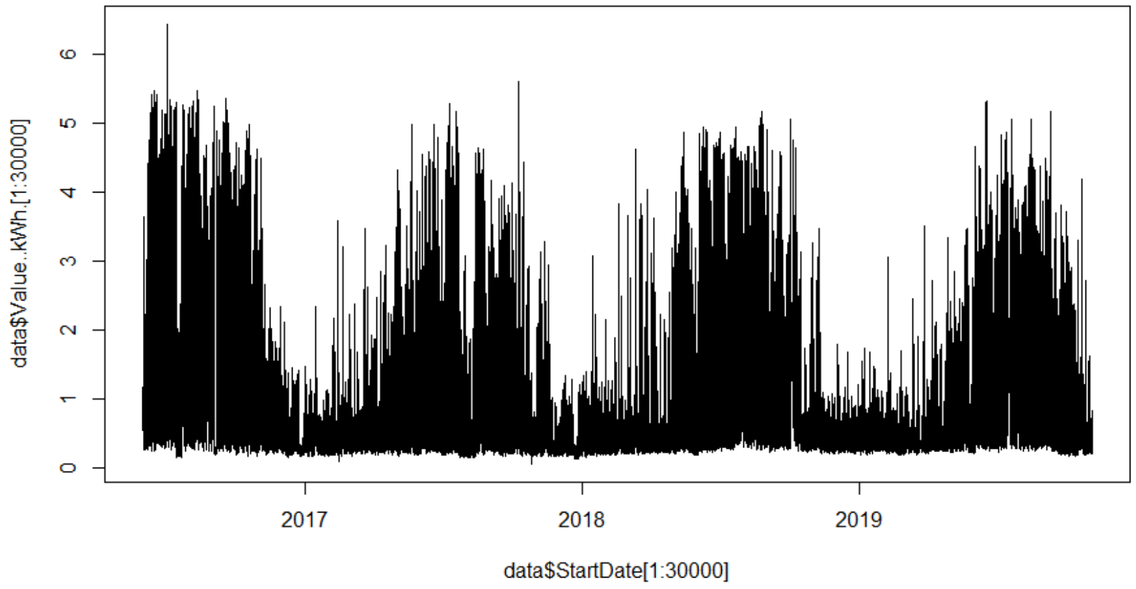


Power usage plots

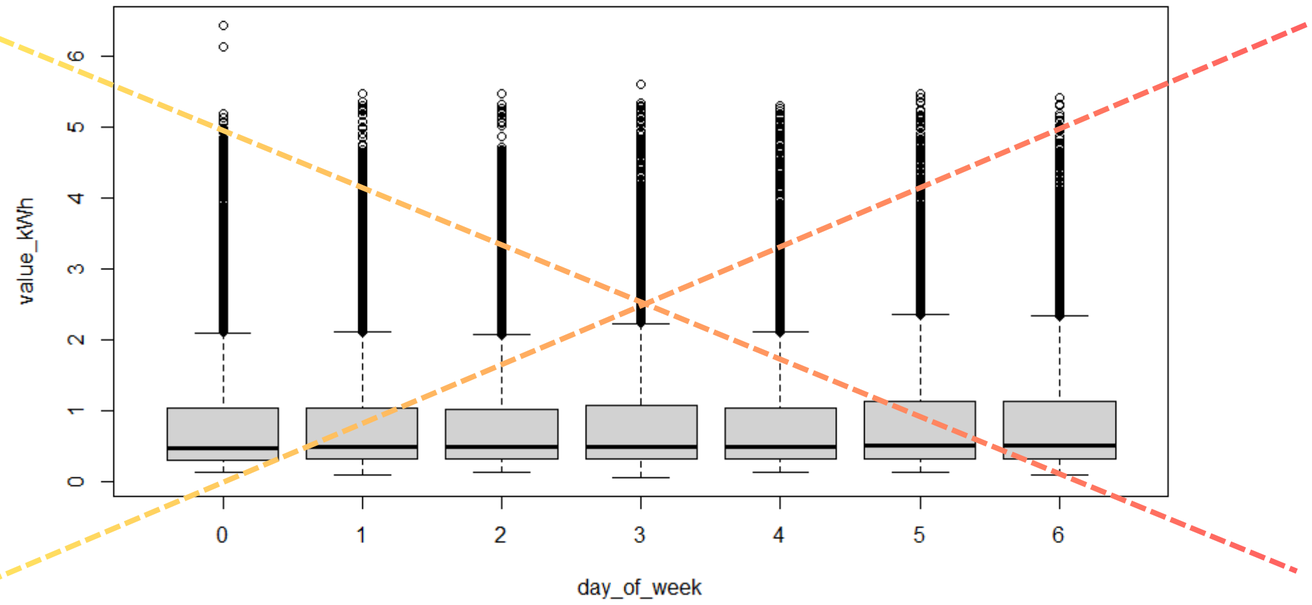


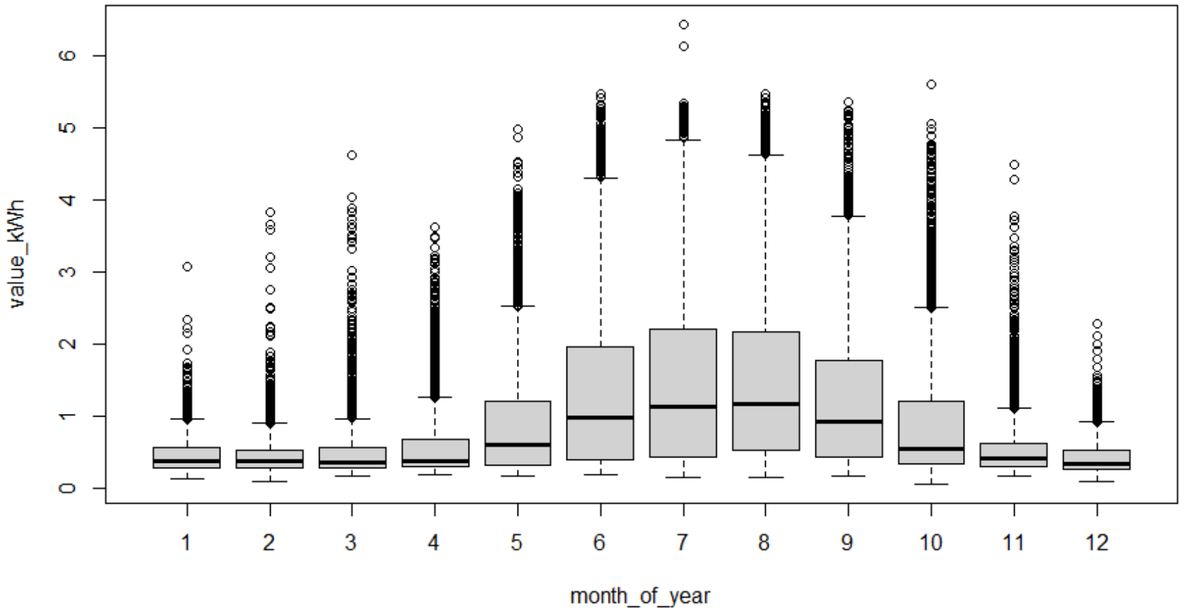
- positive values, continuous, right-skewed data
- strong **hourly** and **seasonal dependences**

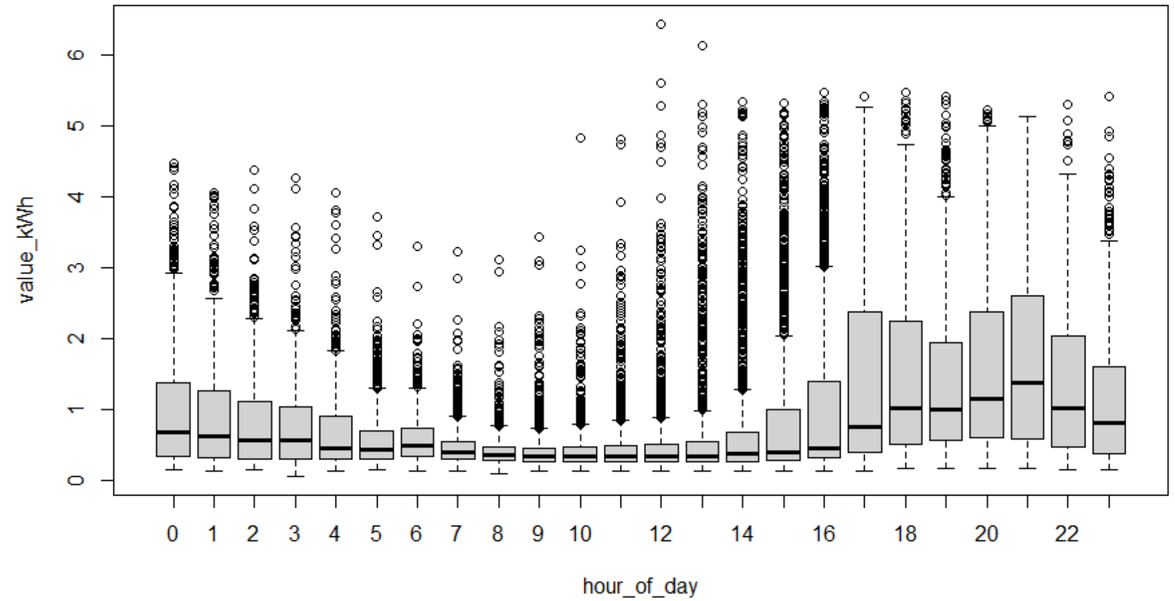


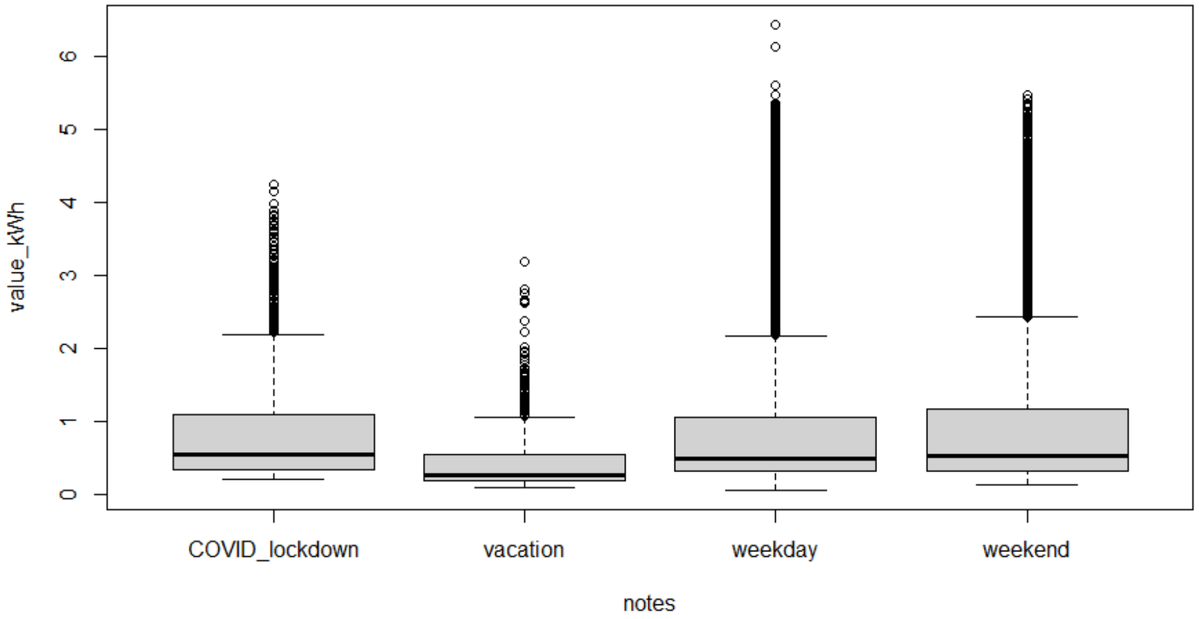


Relationship of power consumption to explanatory variables

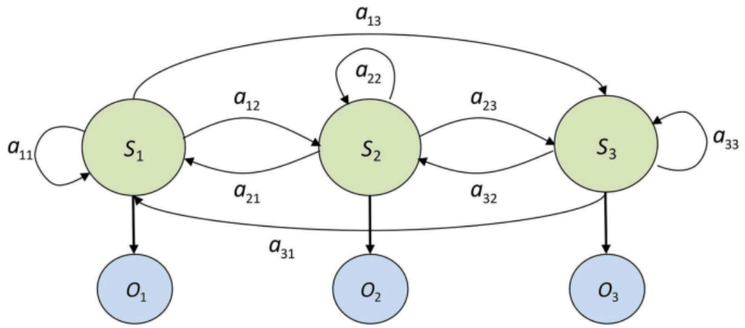






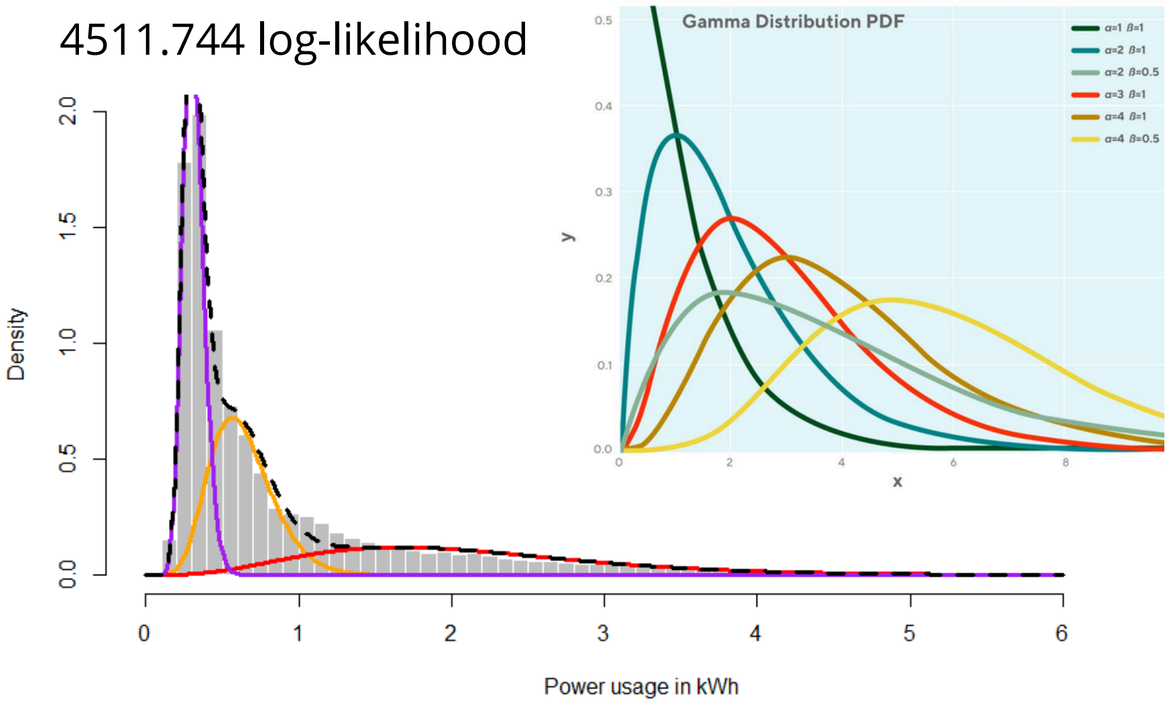


Simple HMM with 3 states



- with **self-written likelihood function**
- **gamma distributions** - parameters and constraints:
  - mu - for each state, positive
  - sigma - for each state, positive
  - Gamma - transition probabilities matrix

4511.744 log-likelihood



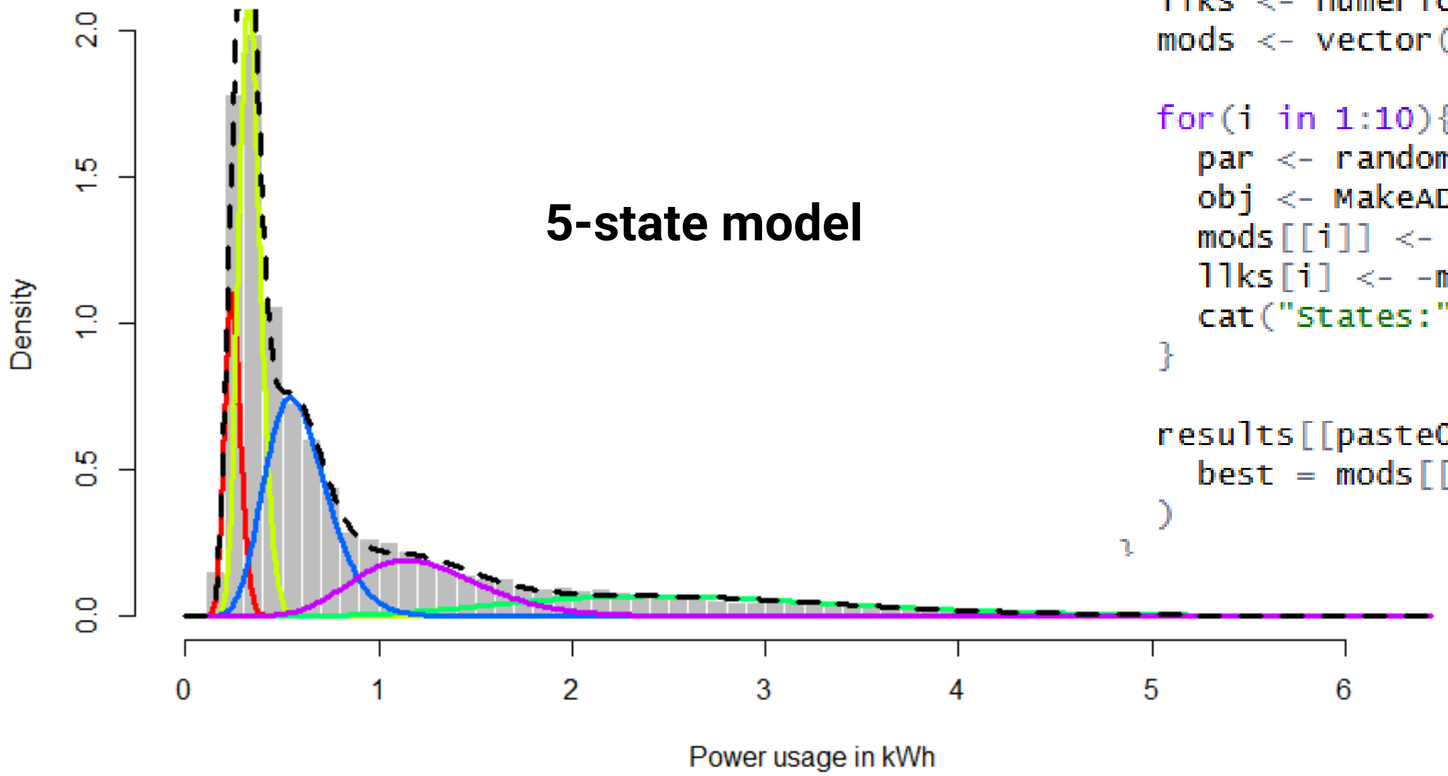

```
llk <- function(theta.star, x) { # unconstrained parameters theta.star
  Gamma <- transform_3x3(theta.star[1:6]) # retransform probabilities - to be
  # retransform mean and standard deviations - to be positive
  mu <- exp(theta.star[7:9])
  sigma <- exp(theta.star[10:12])
  delta <- rep(1 / 3, 3) # 3-states initial probs

  allprobs <- matrix(1, length(x), 3) # allprobs = matrix full of 1s, for ever
  # for t's when allprobs is 1 the diagonal matrix we create in the
  # forward algorithm becomes the identity matrix
  ind <- which(!is.na(x))
  allprobs[ind, ] <- cbind( # now only put density values for non-missing obse
    dgamma(x[ind], shape = mu[1]^2 / sigma[1]^2, scale = sigma[1]^2 / mu[1]),
    dgamma(x[ind], shape = mu[2]^2 / sigma[2]^2, scale = sigma[2]^2 / mu[2]),
    dgamma(x[ind], shape = mu[3]^2 / sigma[3]^2, scale = sigma[3]^2 / mu[3])
  )
  foo <- delta %*% diag(allprobs[1, ]) # alpha_1
  l <- log(sum(foo)) # this will add up to the log-likelihood
  phi <- foo / sum(foo) # standardized alpha_1: phi_1
  for (t in 2:length(x)) {
    foo <- phi %*% Gamma %*% diag(allprobs[t, ]) # phi_t-1 * Gamma * allprobs
    l <- l + log(sum(foo)) # updated log-likelihood
    phi <- foo / sum(foo) # phi_t
  }
  return(-l) # return -llk in order to maximize llk with the minimizer nlm()
}
```

HMMs with 2-5 states

- using **LaMa** package
- 10 randomly initialized models

```
2 states best (min nll) result: 10552.66
3 states best (min nll) result: 4511.837
4 states best (min nll) result: 1757.393
5 states best (min nll) result: 15.44972
```



5-state model

```r
random_par <- function(N, x){ # function to generate starting parameters
  m <- mean(x)
  s <- sd(x)
  list(
    logit_Gamma = rnorm(N * (N - 1), 0, 2), # small values in range -4, +4
                                            # so Gamma has reasonable range of values
    log_mu = log(runif(N, 0.1*m, 2*m)),
    log_sigma = log(runif(N, 0.1*s, 2*s))
  )
}
```

```r
for(N in 2:5){ # for 2-5 number of states
  dat <- list(N = N, d = data$value_kWh)
  llks <- numeric(10) # a vector of length 10
  mods <- vector("list",10)

  for(i in 1:10){ # 10 models, 10 different lists of starting parameters
    par <- random_par(N, data$value_kWh)
    obj <- MakeADFun(nll, par)
    mods[[i]] <- list(opt = nlminb(obj$par, obj$fn, obj$gr), obj = obj)
    llks[i] <- -mods[[i]]$opt$objective # transform negative llk to posi
    cat("States:",N," Run:",i,"\n") # print to console
  }

  results[[paste0("N",N)]] <- list(llks = llks,
    best = mods[[which.max(llks)]] # model with maximum llk
  )
}
```

Model selection w.r.t. the number of states

$$\text{AIC} = -2\log\mathcal{L} + 2 \cdot \#\text{parameters}$$

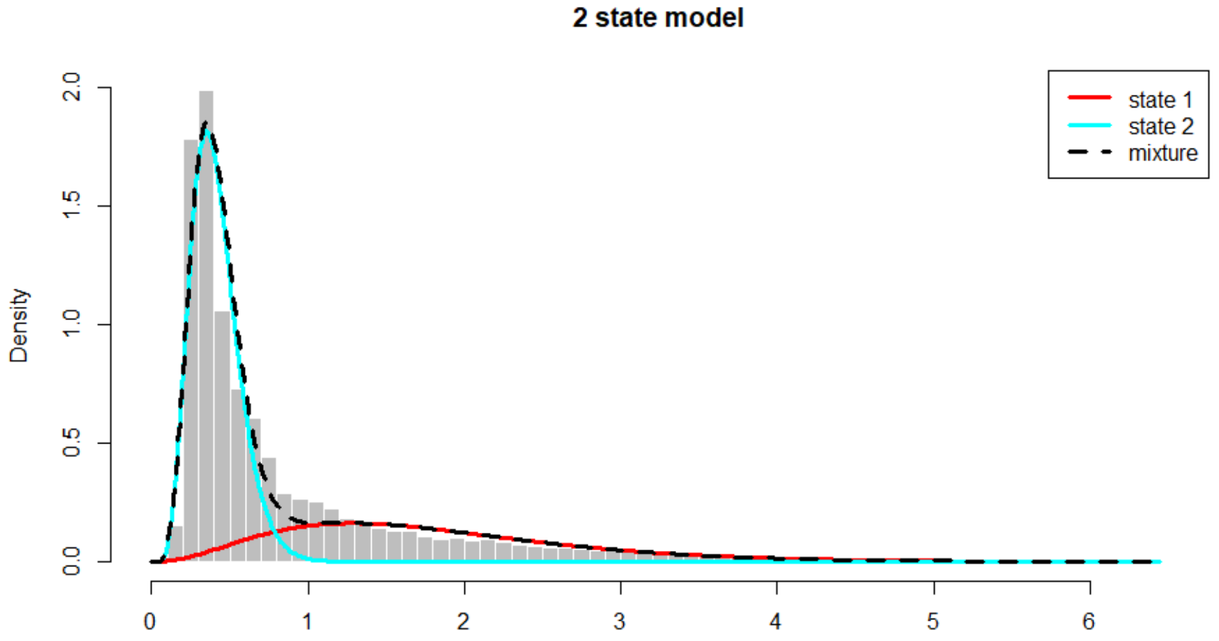
$$\text{BIC} = -2\log\mathcal{L} + \log(T) \cdot \#\text{parameters}$$

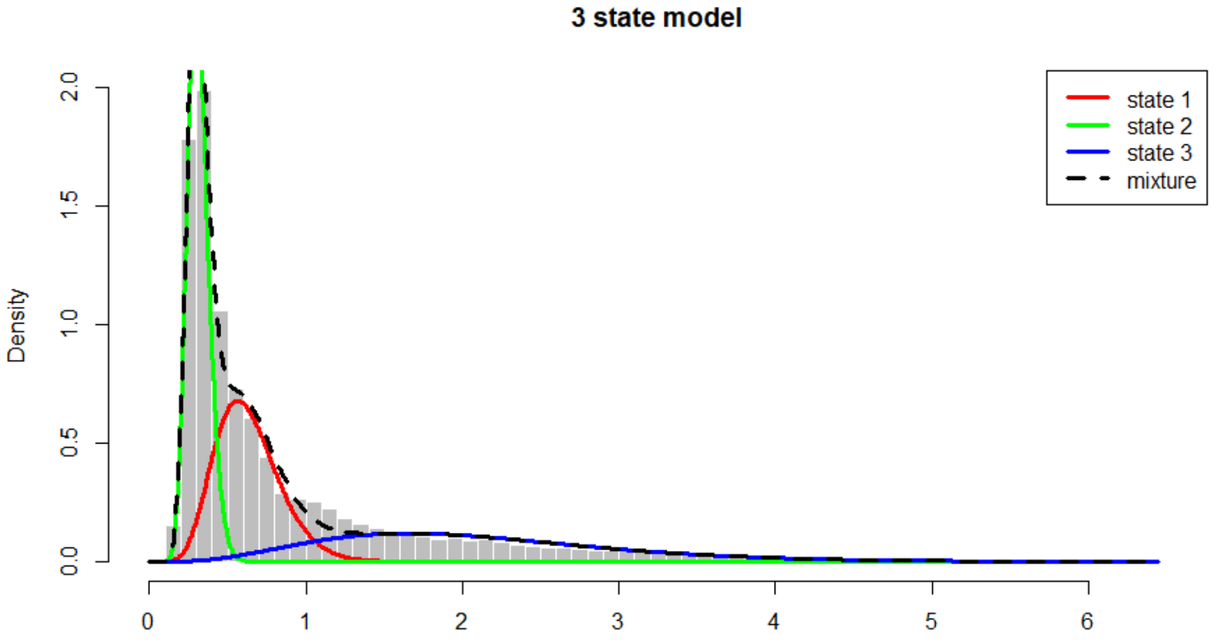
States	LogLik	NumOfParameters	AIC	BIC
2	-10552.65902	6	21117.31803	21168.2577
3	-4511.83713	12	9047.67426	9149.5535
4	-1757.39339	20	3554.78678	3724.5856
5	-15.44972	30	90.89945	345.5976

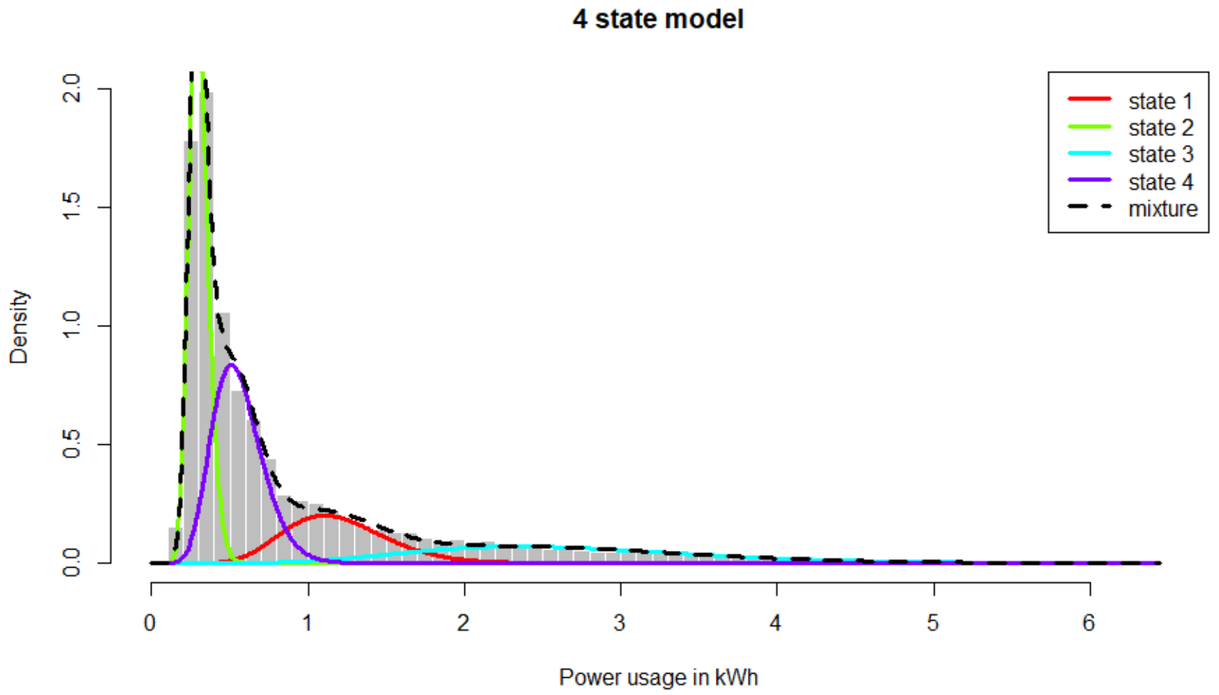
- in both **criteria** we choose the one with the **smallest value**
- (the best models for each number of states were chosen by the llk value)
- best overall: 5-states HMM

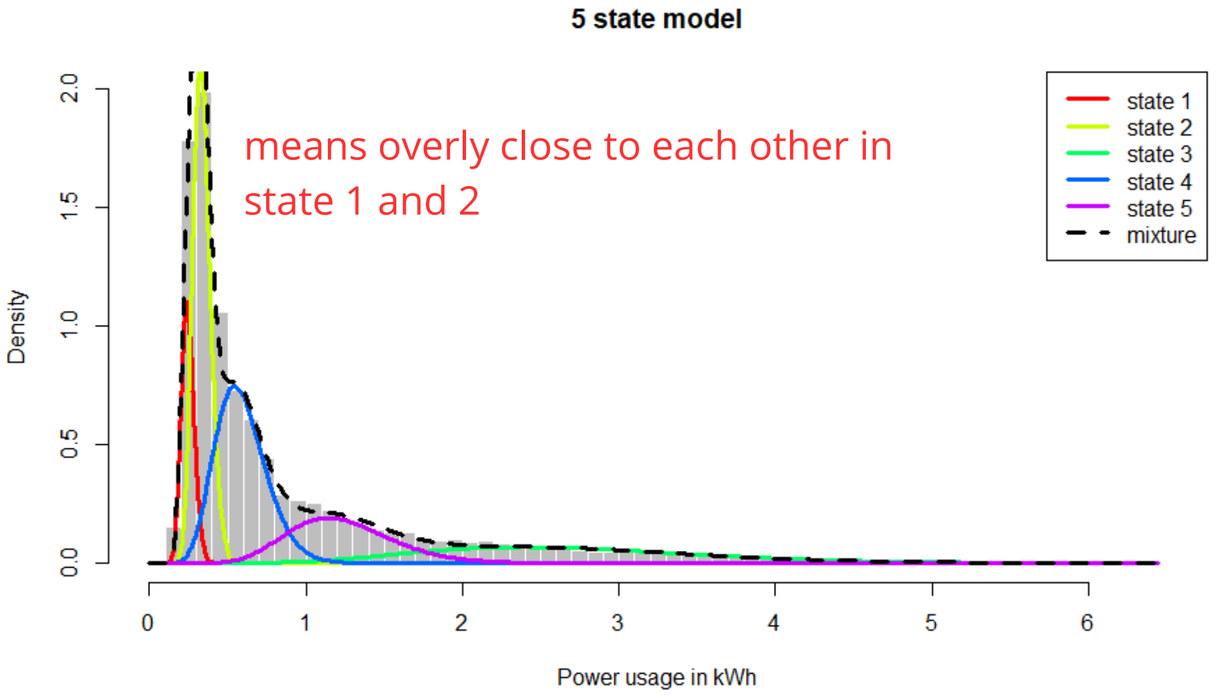
Model checking (on the best ones for each number of states):

- visual fit
- residuals


2 state model
Density
state 1
state 2
mixture


3 state model
Density
state 1
state 2
state 3
mixture

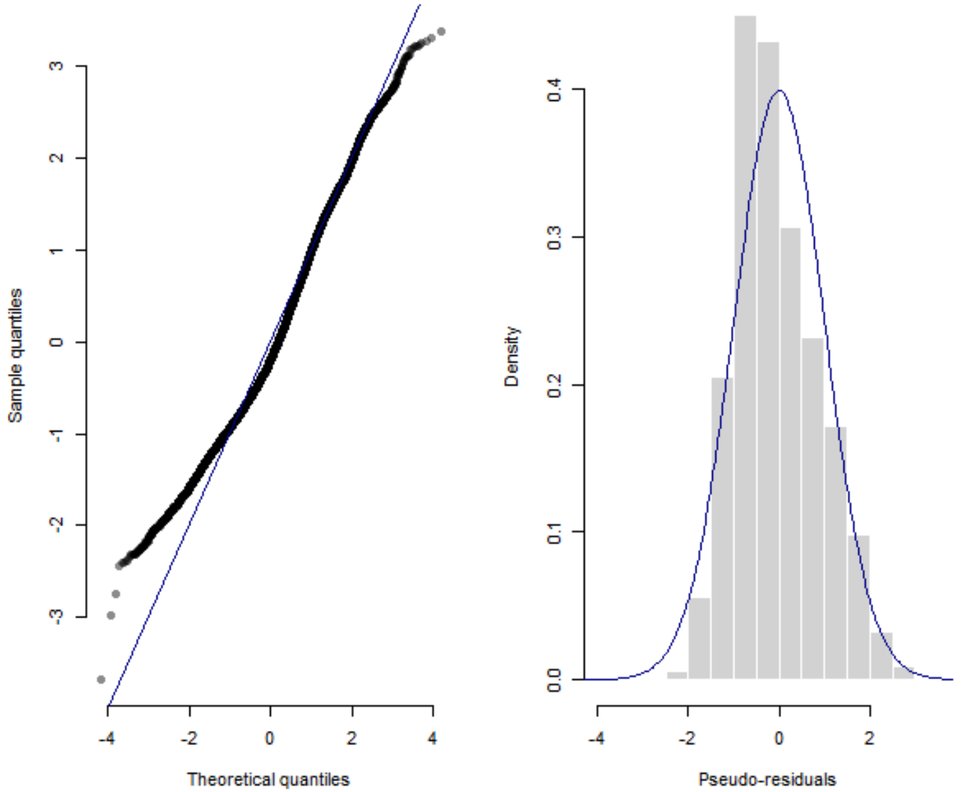

4 state model
Density
Power usage in kWh
state 1
state 2
state 3
state 4
mixture

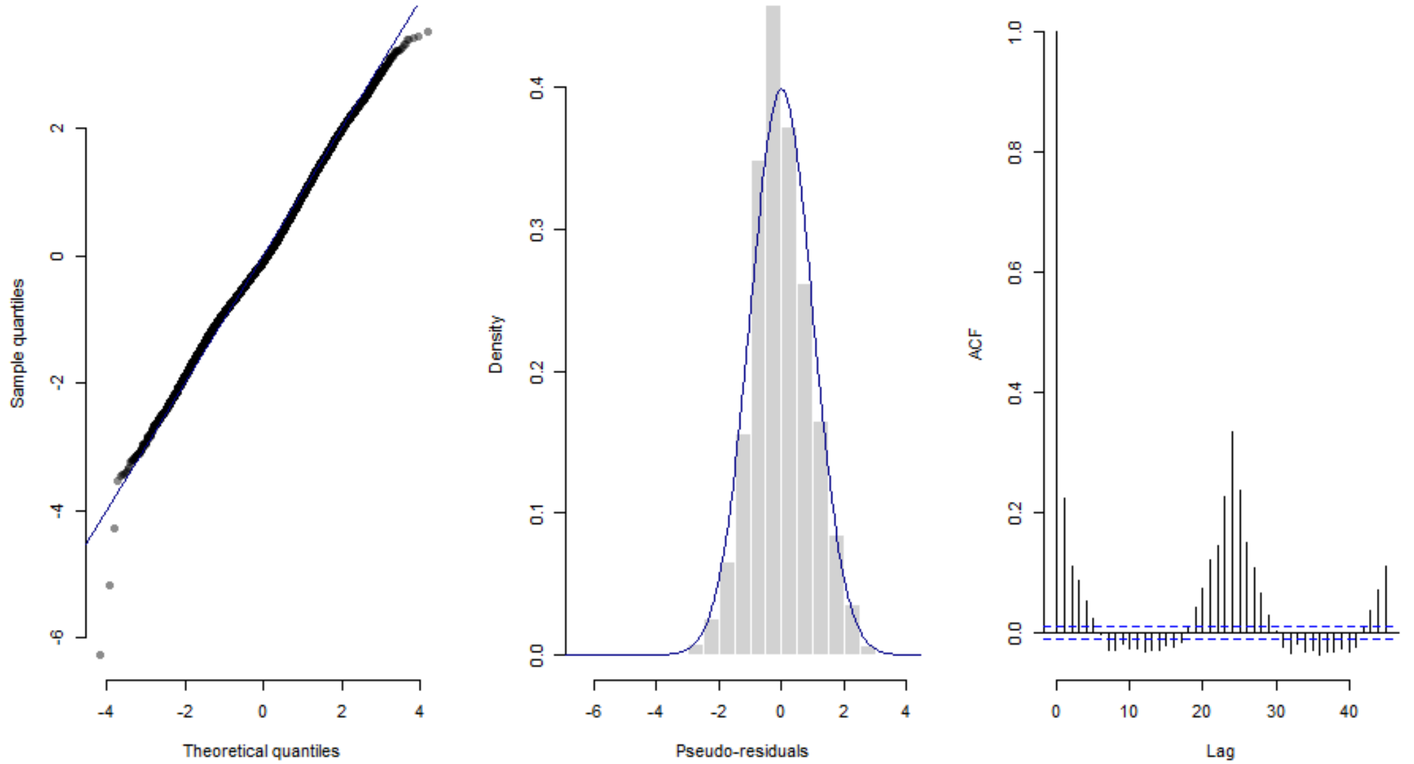

5 state model
means overly close to each other in state 1 and 2
Density
Power usage in kWh
state 1
state 2
state 3
state 4
state 5
mixture

Residuals checking

- **QQ plot** gets better with more states
- **pseudo-residuals** quite normally distributed

2-states HMM





5-states HMM

- **autocorrelation** bad in all
- but gets better with more states

Models with covariates

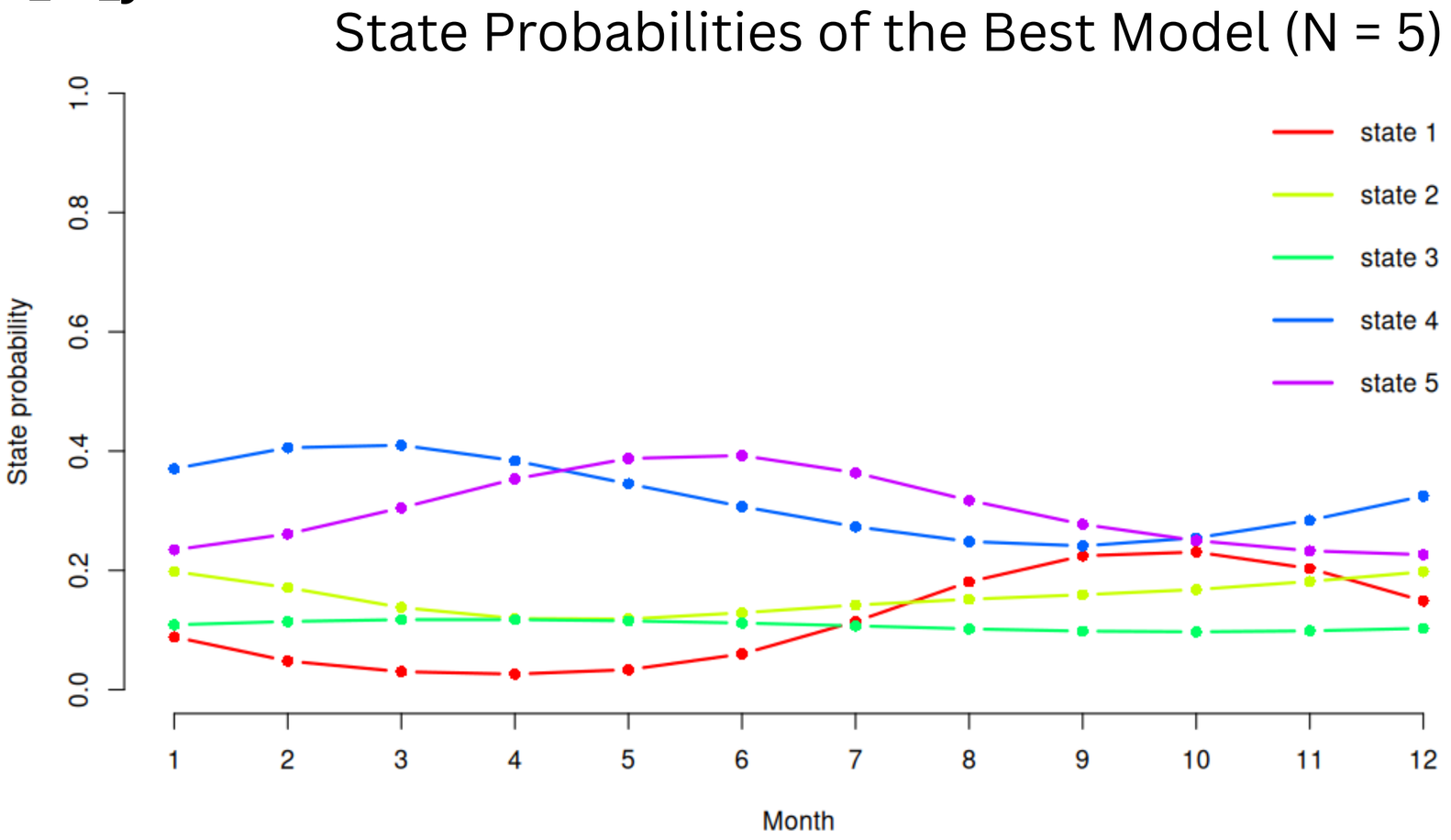
- **month_of_year**

Model selection w.r.t. the number of states

States	LogLik	NumOfParameters	AIC	BIC
2	-10098.8054	10	20217.611	20302.5102
3	-3994.8741	24	8037.748	8241.5068
4	-1180.8233	44	2449.647	2823.2040
-> 5	717.4001	70	-1294.800	-700.5045

Models with covariates

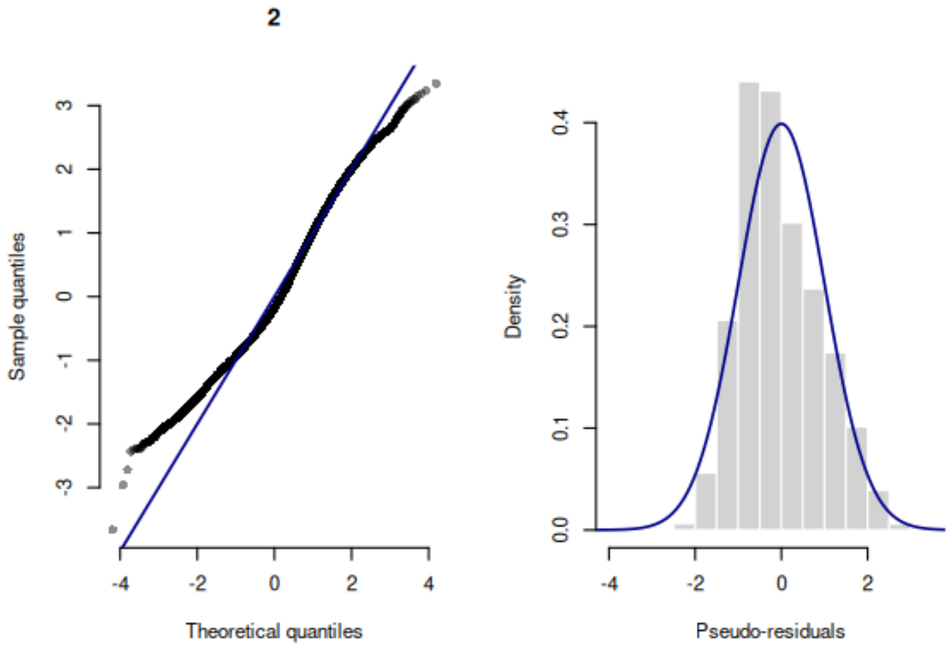
- **month_of_year**

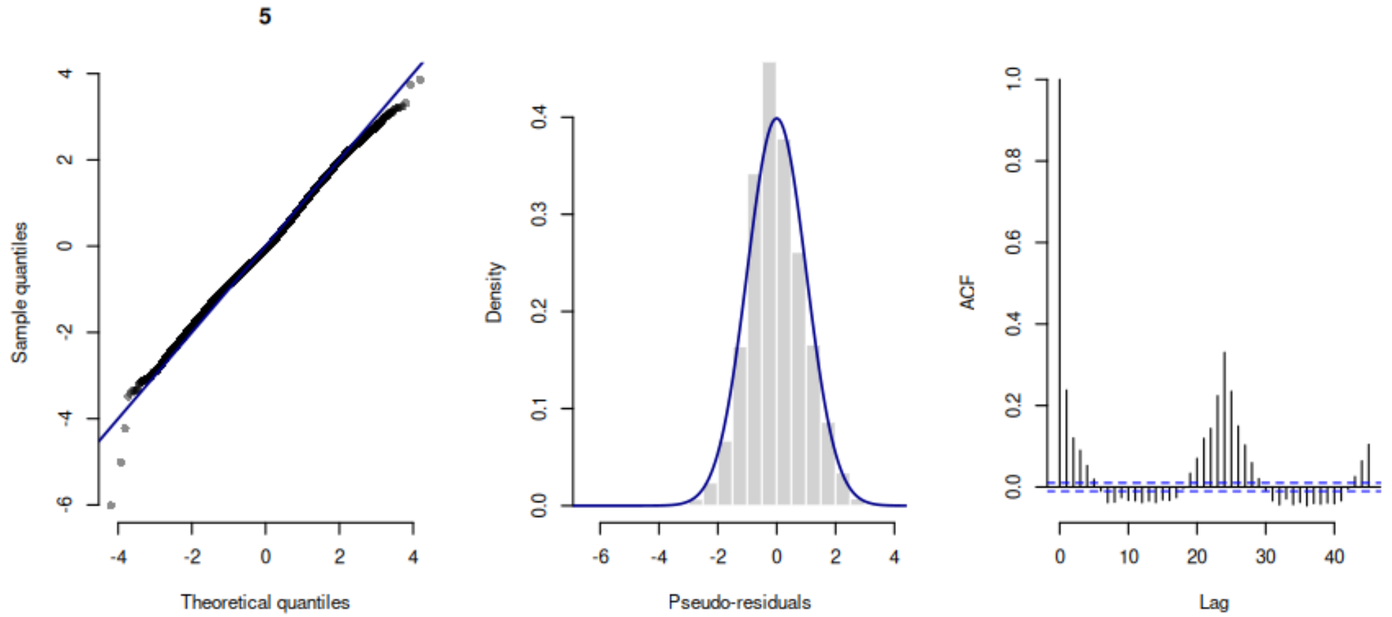


Residuals checking

- **month_of_year**

2-states HMM





5-states HMM

- **autocorrelation** still bad in all

Models with covariates

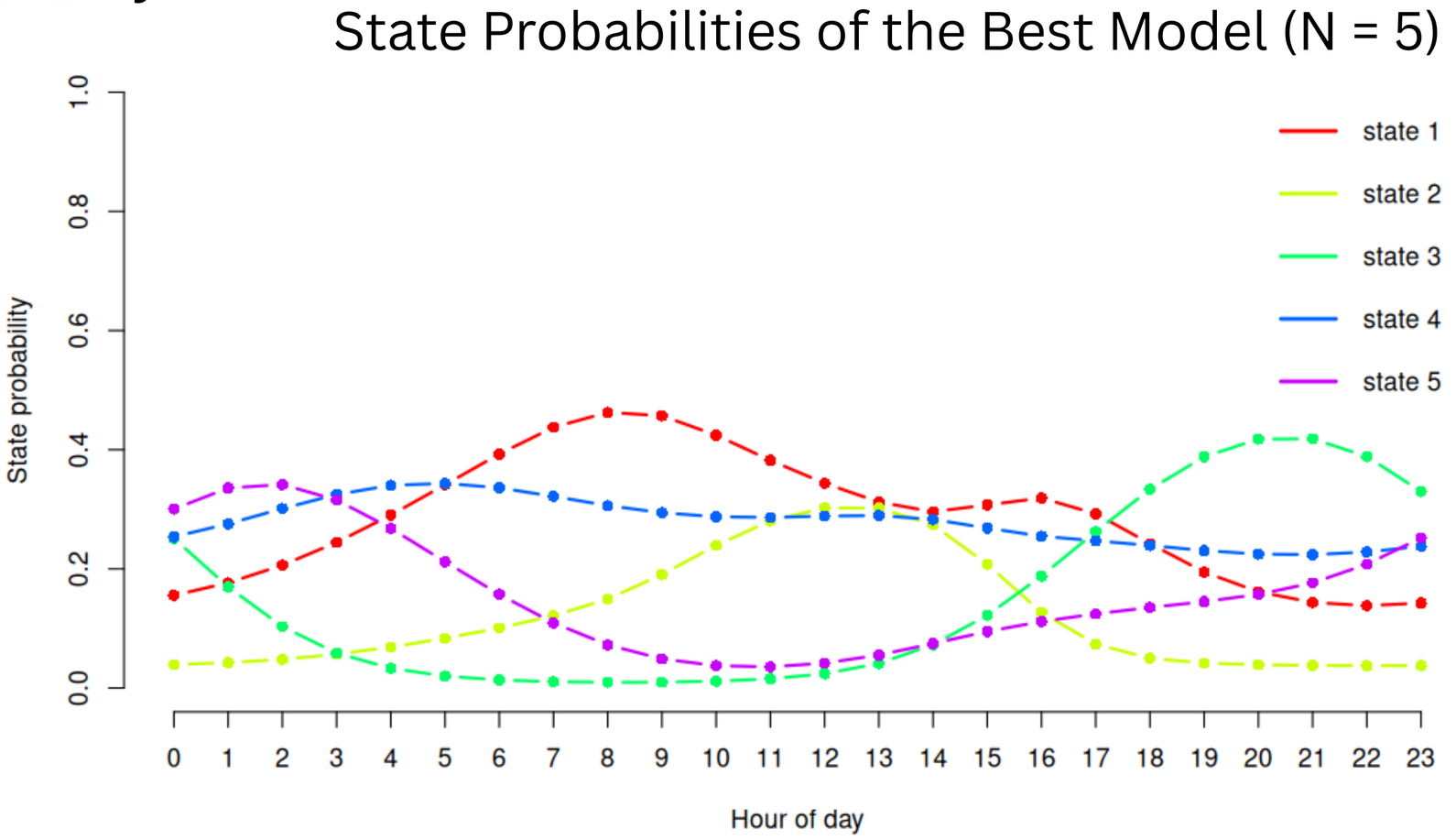
- **hour_of_day**

Model selection w.r.t. the number of states

	States	LogLik	NumOfParameters	AIC	BIC
	2	-9458.0855	10	18936.1711	19021.0705
	3	-3116.9730	24	6281.9460	6485.7045
	4	-104.6508	44	297.3016	670.8589
->	5	1955.4696	70	-3770.9392	-3176.6434

Models with covariates

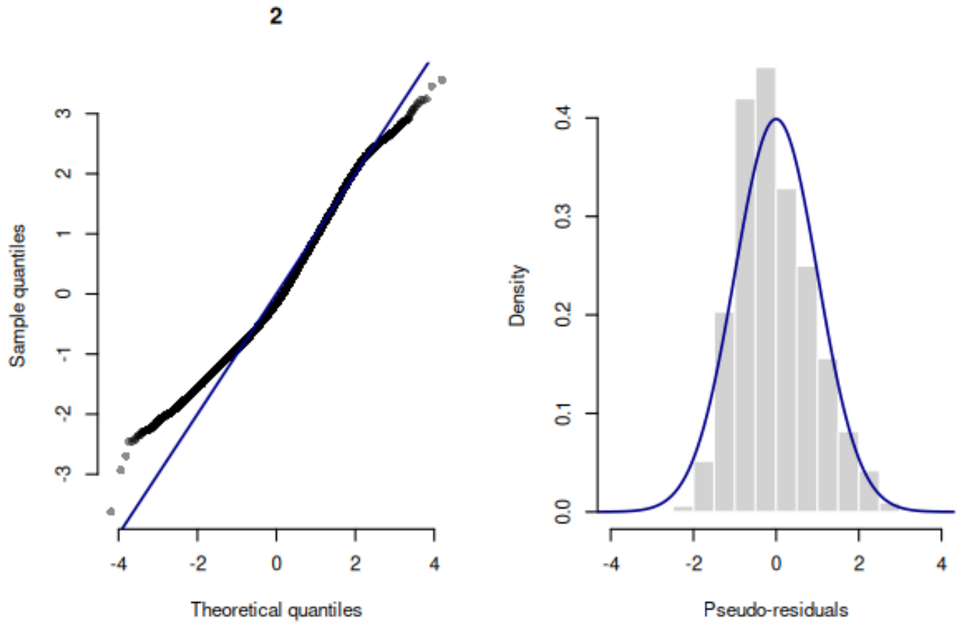
- **hour_of_day**

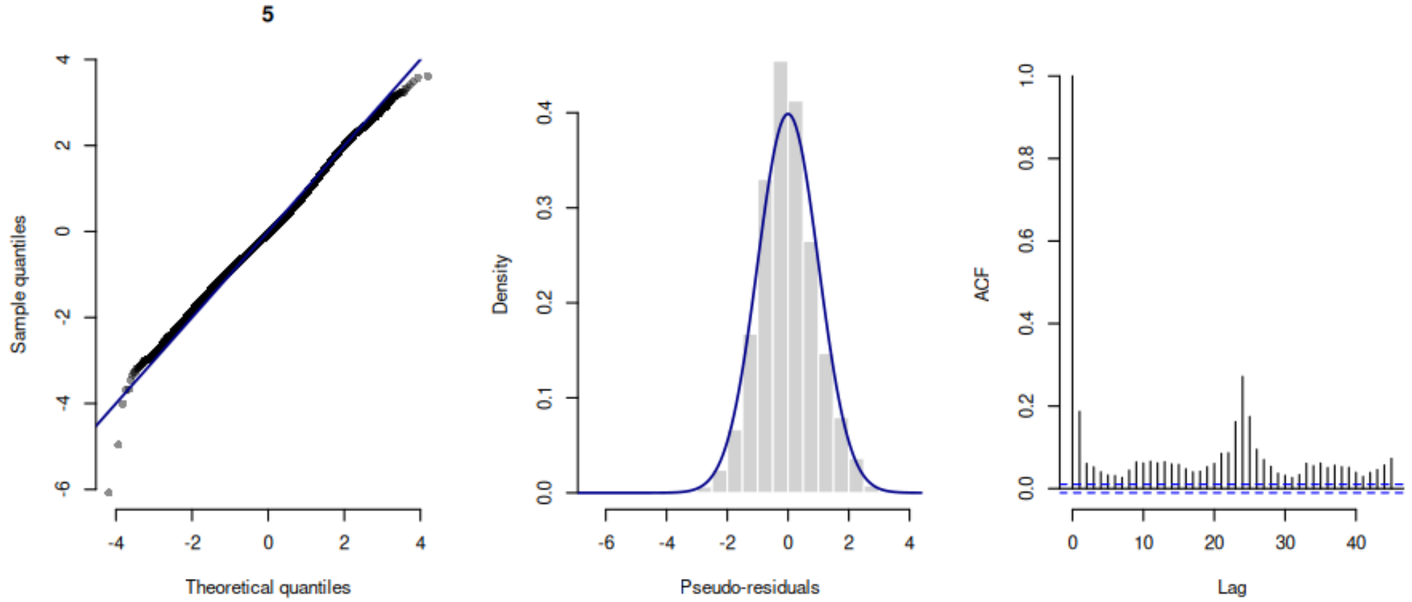


Residuals checking

- **hour_of_day**

2-states HMM





5-states HMM

- **autocorrelation** still bad in all

Model selection (overall)

Number of states
2
3
4
5

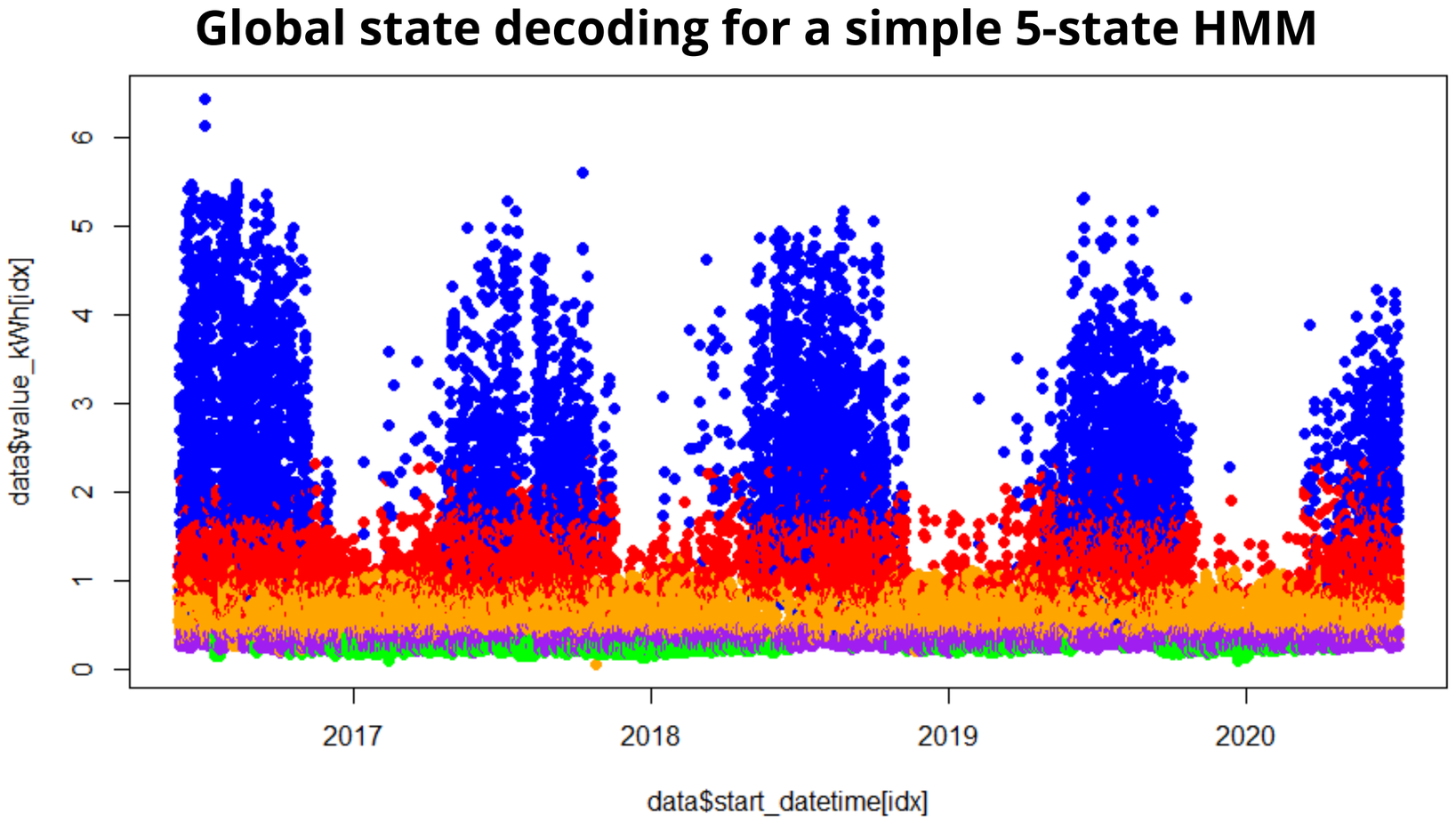
Covariates
-
month_of_year
hour_of_day

Model selection (overall)

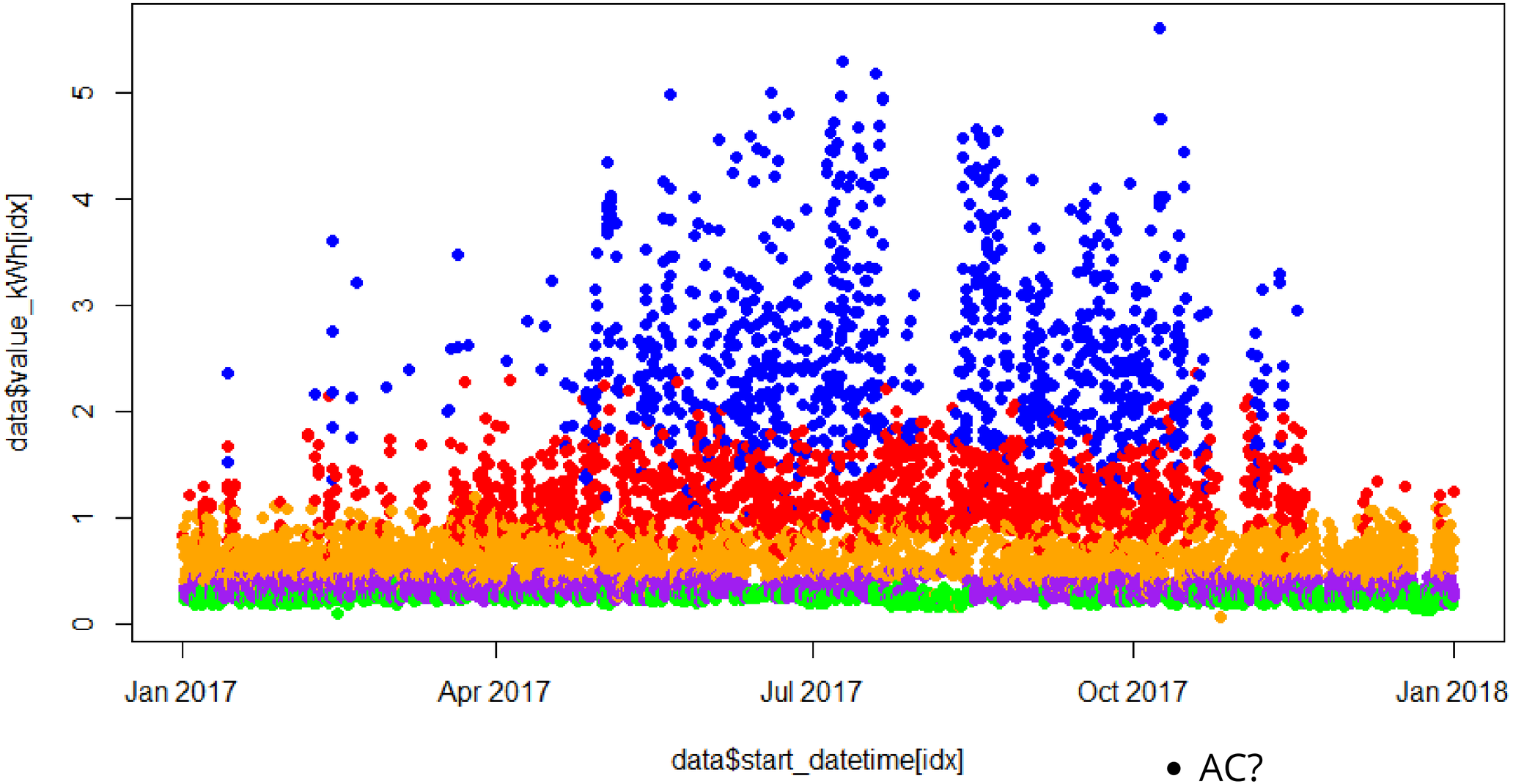
BIC

Number of states	Simple model	Model wrt moy	Model wrt hod
2	*21168.26*	*20302.51*	*19021.07*
3	*9149.55*	*8241.51*	*6485.70*
4	*3724.59*	*2823.20*	*670.86*
5	*345.60*	*-700.50*	***-3176.64***

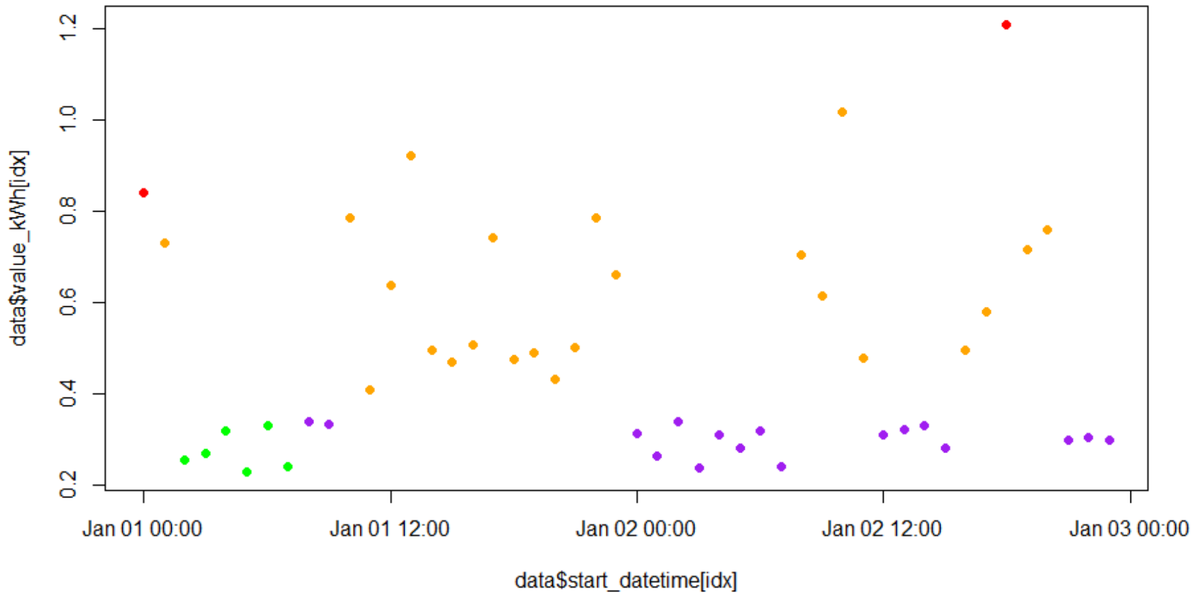
State Decoding



a close-up to 2017



- AC?
- vacations - breaks for state5?

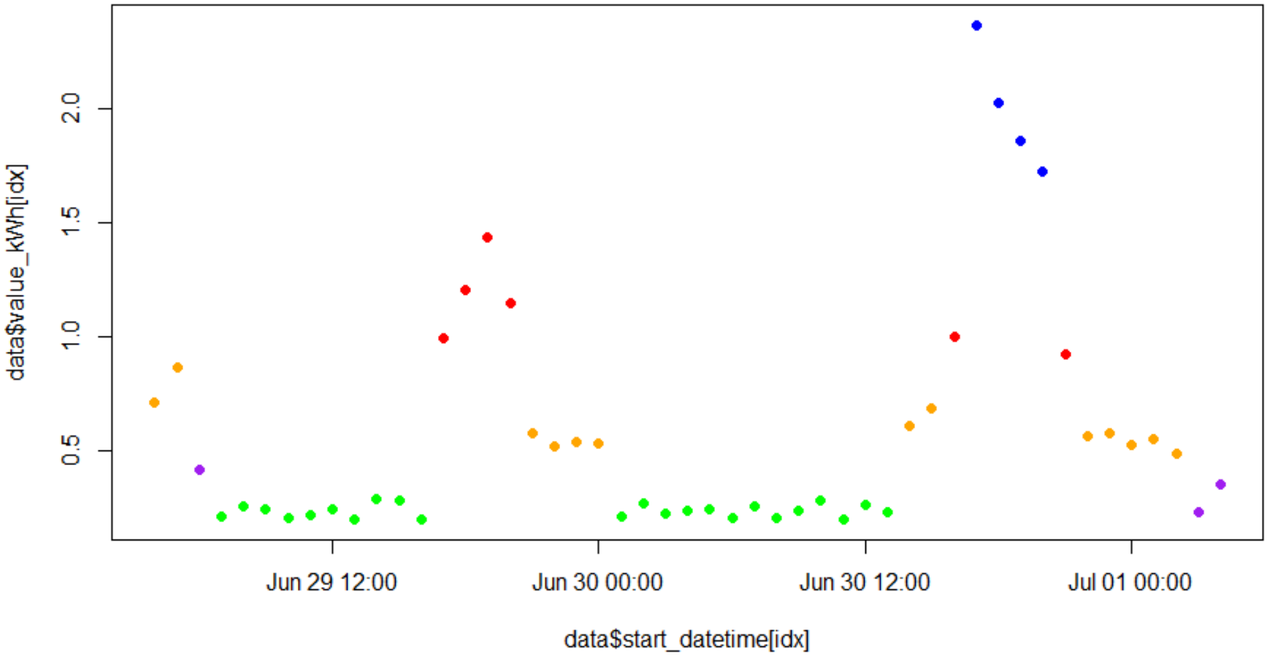


a close-up to see the transitions

col = c("red", "purple", "blue", "orange", "green")
mod$mu # 1.2133267 0.3430480 2.6544948 0.5965250 0.2497793
so: **green, purple, orange, red, blue** would be expected

- usually transitions happen in the expected order – a **state-space model** might be a good idea

- **states 1 & 2** seem problematic – should be merged
- sometimes the process **jumps a state**



Interpretation of the results and reflection

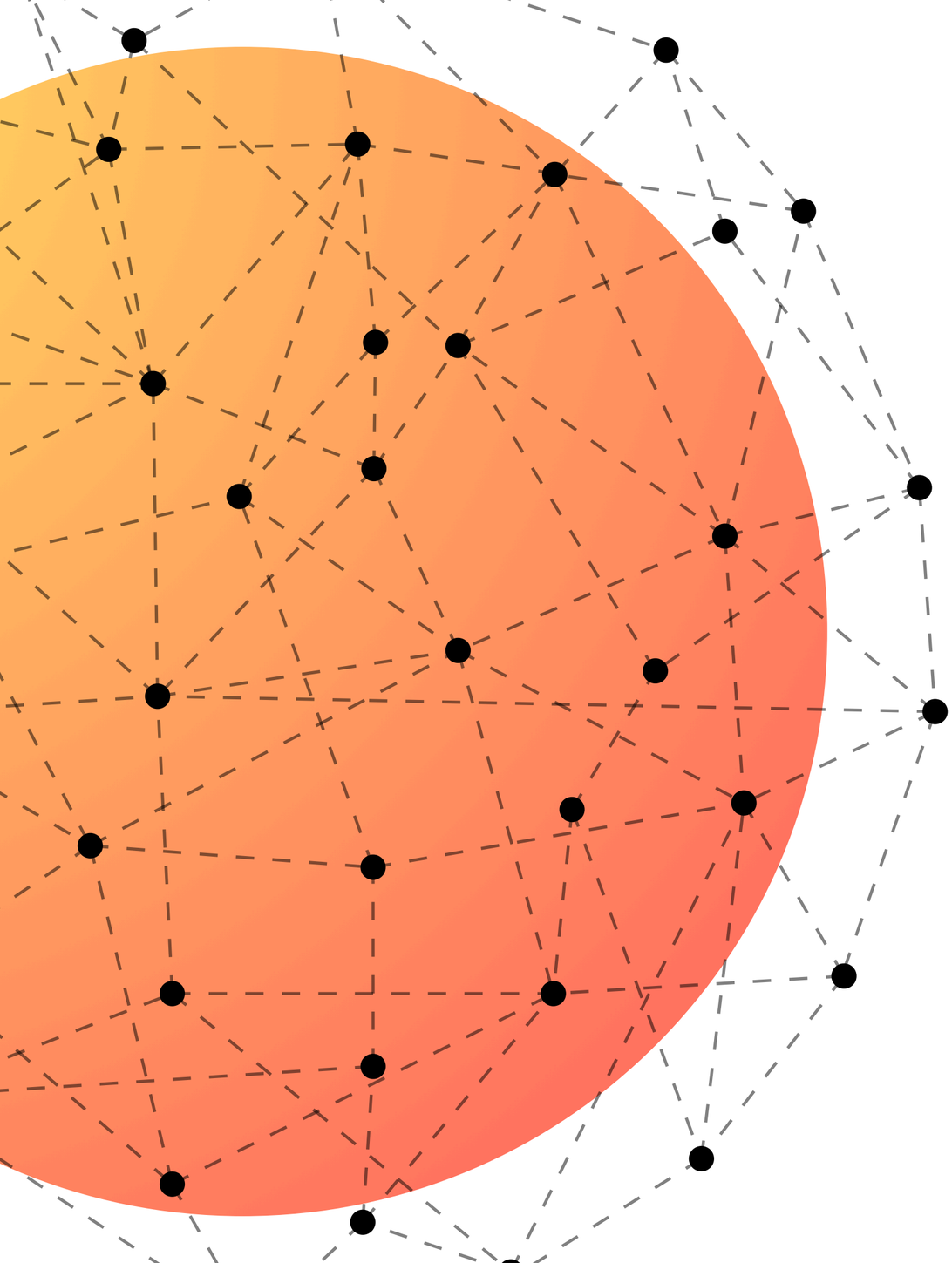
HMM uses dependency on the previous datapoint in time, but maybe the time dependency can be modeled with a **different model more simply** – values are very **cyclical**;
and with better results - especially for **forecasting**, HMMs are only quite good in the first couple of states, then collapse to stationary

Problems:

- model checking – **ACF still** shows a pattern with **large values**
- model selection – **BIC, AIC favour too many states** - overfitting and less interpretable

<u>Possible extensions:</u>

- **extracting the correct DoW**, correcting **“notes”** and using these as **covariates**
- often swiching between states happens in the expected order, might be a good idea to use a **state-space model**



Thank you for your attention